

William Sealy Gosset

A MATH 213 Project

Instructions

Today, we will be discussing the work of the famous statistician William Sealy Gosset. I would like for you all to do some leg-work to find out some facts about his life and his work. Consult the sources below and, with your group, make a slide or two about Gosset. Feel free to include relevant pictures (keep it on topic, please!). If you use sources other than the ones listed below, please cite them. (Posting a link on your slide is fine.)

Suggested sources:

<https://mathshistory.st-andrews.ac.uk/Biographies/Gosset/>

https://en.wikipedia.org/wiki/William_Sealy_Gosset

Two articles posted on blackboard under today's date

Group 2

William Gosset was born in Oxfordshire and he attended university at New College Oxford, where earning degrees in chemistry and mathematics. He is known for things like Student's t-distribution (Student was a pseudonym he went by), the concept of statistical significance, and applications of the Monte Carlo method. Many of these discoveries stem from his work as a statistician for Guinness brewing company - where he was seeking to optimize beer sales and production for the company - and also from his biostatistical research under Karl Pearson at the Galton Eugenics Laboratory in London.

Group 2 slide 2

Notable achievements:

- Invented the T-test, which is a method used to compare the means between two groups.
- Contributed important work on the Poisson limit to the binomial and the sampling distribution of the mean, standard deviation, and correlation coefficient
- Was the head brewer at Guinness brewery



Group 4

- In 1908 William Sealy Gosset, an Englishman publishing under the pseudonym Student, developed the t-test and t distribution.
- he rediscovered the Poisson distribution, which expresses the probability of a given number of events in a fixed amount of time if the events happen with a known constant mean rate and independent from of the time since the previous event
- Gosset obtained a post as a chemist with Arthur Guinness Son and Company in 1899.
- Was friends with two other mathematicians (which was cool because they had big egos)



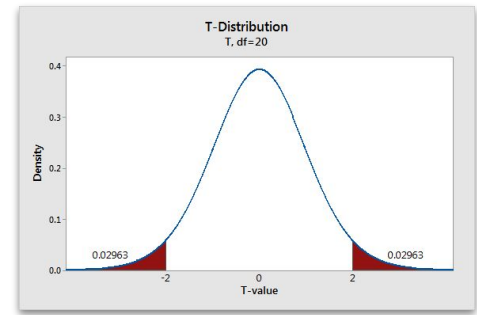
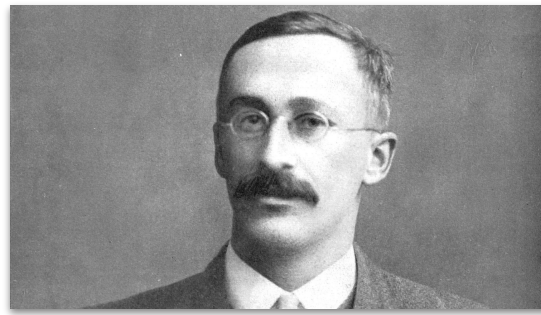
Group 8

- Born June 13th, 1876, and he worked as a Head Brewer for a Guinness factory.
- Did important statistical work while at the dublin factory
- Invented the t-test, handles small samples for quality control in brewing. As well as the t distribution
- Published over 21 academic papers
- Found the tables from arguing with Fisher which the tables were finally published in 1925
- He used the statistics to see which barley grows best with what field and what soil is used

- He developed the **student t-test**, **t-distribution**, and **statistical significance** all to find the best beer
- He wished to deal with the issue of having **small sample sizes** to assistance with beer quality control in brewing
- He rediscovered the **poisson distribution**
- He was an early developer of the proper use of **significance testing** on repeated trials and **decision theory**
- He was known as “Student” because guinness wanted to keep their beer secrets a secret



William Sealy Gosset



MAIN CONTRIBUTION(S):

- ☆ **Invented the t-test** to handle small samples for quality control in brewing
 - Contributed to regression analysis
 - Contributed to studentized residuals
- ☆ “Modern synthesis”
- ☆ Agricultural economics
- ☆ Econometrics

FUN FACTS:

- Got two degrees: one in chemistry and mathematics
- May be more commonly known as “Student”
- He was in an accident in 1934, confined to a bed for 3 months and sustained a limp
- A year later, he took a position as Head Brewer, but passed a year later
- Testimonies describe him as kind, tolerant, modest, gentle, brave – awesome personality on top of being super smart!

Gosset

- Known as “Student” because the brewery did not want to be identifiable or have it known they used these methods, so they made William Gosset choose a pseudonym; he chose “Student” which he chose from his 1906-1907 cell counting notebook, "The Student's Science Notebook".
- Known for Student's t-distribution and test of Statistical significance (t-test)
- T-test is pretty robust due to his preference for a “balanced layout” where treatments and controls are allocated in a balanced fashion
- Really not that important historically as "Fisher would have discovered it all anyway."

William Sealy Gosset was born in June of 1876 in England, with degrees in both chemistry and mathematics, William was a renowned figure in the world of statistics, creating many of the very well known and very heavily used concepts in the field, such as t-distribution and statistical significance, residuals, regression analysis, all under the name Student. The use of this pseudonym was mostly to prevent conflicts that would come from people knowing who he was, thus complicating his work and the work of his boss at the brewery. Most of his work was actually done to help with his work in the brewery, as the t-test was originally created to handle small samples. He was described as a rather modest man, as evidenced by his desire to keep his work more private, rather than take the credit.

Gosset's Children

Mother: Majory Gosset

Harry

- Paediatrician
- First Casualty Officer at Hammersmith Hospital
- Noted for pioneering exchange transfusions for neonates
 - Inventing a simple diagnostic device called the Gosset Icterometer
 - Masterminding the establishment of a new Unit for Premature Babies at Northampton General
 - Named the Gosset Ward in his memory.

Bertha Marian

- Geographer and Nurse

Rush

- Married the Oxford mathematician Douglas Roaf